SIMATS ENGINEERING

Name	Register Number	Course Code	Course
S Susmitha	192571068	ITA0510	Computer Vision

22. Perform Sharpening of Image using Laplacian mask with positive center coefficient.

Mask of Laplacian + addition

$$
\begin{aligned}
g(x,y) &= f(x,y) - [f(x+1,y) + f(x-1,y) \\
&\quad + f(x,y+1) + f(x,y-1) + 4f(x,y)] \\
&= 5f(x,y) - [f(x+1,y) + f(x-1,y) \\
&\quad + f(x,y+1) + f(x,y-1)]
\end{aligned}
$$

0	-1	0
-1	5	-1
0	-1	0

AIM:
Sharpening of Image using Laplacian mask with positive center coefficient.

PROGRAM:

```
import cv2
import numpy as np

image = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png')

if image is None:
    print("Error: Image not found!")
else:
    # Laplacian sharpening kernel
    kernel = np.array([
        [0, 1, 0],
        [1, -8, 1],
        [0, 1, 0]
    ])

    # Apply the sharpening filter
    sharpened = cv2.filter2D(image, -1, kernel)

    # Save the output image
    cv2.imwrite(
        r'C:\Users\Susmitha\Downloads\ITA0510 Lab\Sharpened_Image.jpg',
        sharpened
    )

    # Display the images
    cv2.imshow("Original Image", image)
```

```
cv2.imshow("Sharpened Image", sharpened)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT:

OUTPUT:

RESULT:

Thus the program is executed successfully.